Solution

If the given matrix M is singular, then

$$\begin{vmatrix} x & y \\ z & 1 \end{vmatrix} = 0.$$

That is, $x - yz = 0$.

Hence the possible ways of selecting (x, y, z) are

$$\{(1,1,1),(2,1,2),(2,2,1),(3,1,3),(3,3,1)\} = A(say)$$

The number of favourable cases $n(A) = 5$

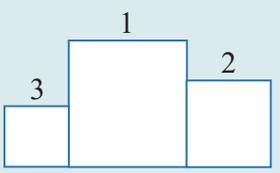
The total number of cases are $n(S) = 3^3 = 27$

The probability of the given matrix is a singular matrix is

$$p = \frac{n(A)}{n(S)} = \frac{5}{27}.$$

Example 12.10

For a sports meet, a winners' stand comprising of three wooden blocks is in the form as shown in figure. There are six different colours available to choose from and three of the wooden blocks is to be painted such that no two of them has the same colour. Find the probability that the smallest block is to be painted in red, where red is one of the six colours.



Solution

Let S be the sample space and A be the event that the smallest block is to be painted in red.

$$n(S) = {}^6P_3 = 6 \times 5 \times 4 = 120$$

$$n(A) = 5 \times 4 = 20$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{20}{120} = \frac{1}{6}.$$

	6	6	3
$n(S)$	6	5	4
$n(A)$	5	4	Red

12.4.3 ODDS

The word **odds** is frequently used in probability and statistics. Odds relate the chances in favour of an event A to the chances against it. Suppose 'a' represents the number of ways that an event can occur and 'b' represents the number of ways that the event can fail to occur.

The odds of an event A are $a : b$ in favour of an event and

$$P(A) = \frac{a}{a+b}.$$

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